

Network Intrusion Detection using Artificial Intelligence Techniques

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Abstract

Network security is becoming very important because cyber attacks are increasing day by day. In this project, we implemented different Artificial Intelligence and Machine Learning techniques for detecting attacks in network traffic. The dataset used was `network_traffic.csv`. We performed data exploration, implemented a Simple Reflex Agent, K-Nearest Neighbors (KNN), Naive Bayes, Logistic Regression, K-Means Clustering, and Genetic Algorithm. The results showed that AI techniques can help in identifying suspicious network activities and improve security.

1. Introduction

A Network Intrusion Detection System (NIDS) is a system that monitors network traffic and identifies unusual or harmful activities. It helps organizations protect their data and systems from attackers.

Machine Learning and Artificial Intelligence are commonly used in intrusion detection because they can analyze large amounts of data and find patterns that indicate attacks. In this project, different AI approaches were applied on a network traffic dataset to understand how they work and compare their results.

The main objective of this project was to implement classical AI and machine learning techniques for intrusion detection and analyze their performance.

2. Dataset Description

The dataset used in this project was **`network_traffic.csv`**.

Dataset Characteristics:

- Total Records: 6000

- Features: 15 numerical features
- Target Variable: label
- Label 0 = Normal Traffic
- Label 1 = Attack Traffic

The dataset was already cleaned and contained only numerical values. Therefore, no encoding was required before applying machine learning algorithms.

3. Task 1 – Data Exploration

In this task, the dataset was explored using Python libraries such as Pandas and Matplotlib.

The following operations were performed:

- Displayed first few rows of dataset
- Checked dataset shape
- Generated summary statistics
- Created class distribution graph
- Generated histograms for different features

From the exploration, it was observed that the dataset contains both normal and attack traffic records. The visualizations helped in understanding feature distributions and identifying important characteristics of network traffic.

(Insert Dataset Shape Screenshot Here)

(Insert Class Distribution Graph Here)

(Insert Histograms Here)

4. Task 2 – Simple Reflex Agent

A Simple Reflex Agent was implemented using predefined rules.

Rules Used:

- If `error_rate > 0.5`, classify as Attack.
- If `same_srv_rate < 0.5`, classify as Attack.
- Otherwise classify as Normal.

The agent makes decisions only on the current input and does not learn from previous experiences.

Result

Accuracy obtained:

Accuracy = 0.8573 (85.73%)

This result shows that simple rule-based systems can achieve good performance, but they are limited because they cannot learn from data.

5. Task 3 – Supervised Learning Algorithms

5.1 K-Nearest Neighbors (KNN)

KNN classifies a data point based on the majority class among its nearest neighbors. Euclidean Distance was used to calculate similarity between data points.

Result

Prediction = 0

Actual = 0

The predicted label matched the actual label, showing that the algorithm correctly classified the tested sample.

5.2 Naive Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes Theorem. It assumes that features are independent from each other.

Result

Prior Normal = 0.5

Prior Attack = 0.5

Feature Counting Complete

The results indicate that the dataset is balanced because both classes have equal prior probabilities.

5.3 Logistic Regression

Logistic Regression is a binary classification algorithm that predicts probabilities using the sigmoid function.

The dataset was standardized before training. Gradient Descent was used to optimize weights and bias.

Result

Training Complete

Final Weights:

[0.33458056, 0.12203138, 0.00836804, 0.52286797, 0.34409022, -1.17016392, -0.04111806, 1.00169219, -0.14657516, 1.15423939, -1.31241076, -0.25355406, -0.52363742, -1.27798003, 0.65787315]

Final Bias:

0.40733742073190443

The trained model successfully learned weights for all features. Positive and negative weights indicate how each feature affects classification.

6. Task 4 – K-Means Clustering

K-Means is an unsupervised learning algorithm that groups similar data points together.

In this project:

- Number of Clusters (k) = 2
- Euclidean Distance was used
- Centroids were updated iteratively until convergence

Result

Final Centroid 1:

[2530.7619, 4012930.9048, 0.0, 0.0, 1.0, 0.8095, 0.0, 1.4762, 1.4286, 0.0476, 0.9843, 0.0319, 90.1429, 53.5238, 0.5695]

Final Centroid 2:

[406.4981, 1994.7891, 2962.1074, 0.0284, 0.2034, 0.3835, 0.1803, 80.7118, 27.6642, 0.2944, 0.6571, 0.0608, 180.9612, 112.6399, 0.5190]

The algorithm successfully divided the dataset into two clusters representing different traffic behaviors.

7. Task 5 – Genetic Algorithm

Genetic Algorithm is an optimization technique inspired by natural evolution.

Parameters Used:

- Chromosome Length = 15
- Population Size = 20
- Generations = 50
- Mutation Rate = 0.05
- Crossover Rate = 0.8

Operations Performed:

- Selection
- Crossover
- Mutation

Result

The fitness value improved during evolution.

Highest Fitness Achieved:

Fitness = 14

Generation 28 and Generation 29 achieved the highest fitness score.

This shows that the Genetic Algorithm successfully improved solutions over generations and found near-optimal chromosomes.

8. Discussion

The project demonstrated different Artificial Intelligence approaches for intrusion detection.

The Reflex Agent achieved an accuracy of 85.73%, which is good considering it uses only predefined rules. KNN successfully classified the tested sample correctly. Naive Bayes showed balanced class probabilities and prepared the data for probabilistic classification.

Logistic Regression learned meaningful feature weights through gradient descent and provided a mathematical approach to classification. K-Means successfully grouped network traffic into two clusters without using labels. The Genetic Algorithm improved fitness values over generations and demonstrated optimization using evolutionary techniques.

Overall, machine learning methods were more flexible than rule-based methods because they learn patterns directly from data.

9. Conclusion

In this project, multiple AI and Machine Learning techniques were applied for Network Intrusion Detection. Data exploration helped understand the dataset, while supervised and unsupervised learning methods were used for classification and clustering. The Genetic Algorithm demonstrated optimization capabilities. The project provided practical understanding of how Artificial Intelligence can be used to improve network security and detect malicious activities.

Statement of Contribution

The work was divided among both team members as follows:

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- Task 1: Data Exploration
- Task 2: Simple Reflex Agent
- Task 3A: K-Nearest Neighbors (KNN)
- Task 3B: Naive Bayes

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- Task 3C: Logistic Regression
- Task 4: K-Means Clustering
- Task 5: Genetic Algorithm

Both members contributed in testing, debugging, discussion of results, and final report preparation.